Fiscal Year:	FY 2026	**Task Last Updated:**	FY 11/21/2025
PI Name:	Wood, Scott J. Ph.D.		
Project Title:	Sensorimotor Predictors of Postlanding Functional Task Performance		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: PI returned to NASA JSC in January 2017. PI was at Azusa Pacific University from August 2013 – January 2017; prior to August 2013, PI was at NASA JSC.		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2019	**End Date:**	05/30/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/30/2024 per the PI (Ed., 3/3/25) NOTE: End date changed to 09/30/2024 per C. Ribeiro/HHC (Ed., 10/2/23) NOTE: End date changed to 09/30/2023 per PI (Ed., 7/7/21)		
Key Personnel Changes/Previous PI:	Drs. Bloomberg, Reschke and Feiveson have retired. Dr. Bloomberg continued as an unpaid collaborator. Per the Principal Investigator, Timothy Macaulay joined the project team as well (Ed., 1/3/26).		
COI Name (Institution):	Bloomberg, Jacob Ph.D. (NASA Johnson Space Center--Emeritus) Clark, Torin Ph.D. (University of Colorado Boulder, Aerospace Engineering Sciences) Koppelmans, Vincent Ph.D. (University of Utah) Oddsson, Lars Ph.D. (University of Minnesota) Peters, Brian Ph.D. (KBR/NASA Johnson Space Center) Seidler, Rachael Ph.D. (University of Florida) Theriot, Corey Ph.D. (University of Texas Medical Branch) Macaulay, Timothy (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Spaceflight drives adaptive changes in healthy individuals appropriate for sensorimotor function in a microgravity environment. These changes are maladaptive for return to earth's gravity. The inter-individual variability of sensorimotor decrements is striking, although poorly understood. The goal of this study is to identify a set of behavioral, neuroimaging and genetic measures that can be used to predict early post-flight performance on a set of sensorimotor tasks. Astronauts are being recruited who previously participated in sensorimotor field tests and/or posturography soon after long-duration spaceflight. Behavioral tests include assessments of sensory dependency and adaptability. Visual dependency involves treadmill walking while viewing a moving virtual visual scene. Vestibular sensitivity is measured while seated with eyes closed during lateral translations. Proprioception dependency is measured during one-legged stance with eyes closed on a horizontal air-bearing surface. Ground assessment of adaptability is performed (1) during treadmill walking with a virtual linear hallway and a moving walking surface, and (2) during multiple trials of navigating an obstacle course while wearing reversing prisms. The neuroimaging tests will characterize individual differences in regional brain volumes (using Structural MRI) and white matter microstructure (using Diffusion Tensor Imaging) to serve as potential predictors of adaptive capacity. The genetic tests will utilize saliva samples to examine variations in four genes chosen because of their ability to differentiate sensorimotor adaptation ability in a normative population, including Catechol-O-methyltransferase (COMT), Dopamine Receptor D2 (DRD2), Brain-derived neurotrophic factor (BDNF) and the a2-adrenergic receptor. A subjective survey was introduced to obtain ordinal measures of how sensorimotor adaptation impacted tasks both early inflight and postflight, and the general timeframe that movements were restricted. This study will utilize data (n=6 astronauts) from a previous Predictors study (PI Mulavara, grant NCC 9-58-SA03801). Deliverables: Statistical models will use combinations of behavioral metrics, brain structure metrics and genomic polymorphisms to understand individual decrements in post-flight functional task outcomes. We expect that understanding the relationships between these sensorimotor biomarkers and post-flight functional task performance will improve both our understanding of the individual variability and our strategy to optimize sensorimotor countermeasures. Study Participants: This study will target recruitment of up to 30 astronauts who previously flew on ISS, including 6 obtained via the original Predictors study. Risk Characterization, Quantification, Evidence: This project was directed toward the risk titled “Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks.” This study is addressing gaps SM-101 (risk characterization of postural control and locomotion) and SM-104 (evaluate weightlessness-induced changes in sensorimotor/vestibular function with changes in other brain functions). This study will help quantify inter-individual variability of sensorimotor decrements. Countermeasure, Prototype Hardware or Software: The information derived from this study may be applied towards optimizing countermeasures based on preflight tests of sensorimotor biomarkers.
Rationale for HRP Directed Research:	The proposed works qualifies for directed research under the "Highly Constrained Research" category in the Human Research Program (HRP) Unique Processes, Criteria, and Guidelines. This project will utilize the findings from two previous studies (SM Predictors-Ground, SM Predictors-Retrospective) that were funded by the NASA Human Research Program (HRP) through a National Space Biomedical Research Institute (NSBRI) cooperative agreement. These studies have validated, in a non-astronaut ground population, a unique set of measures that were predictive of adaptation in response to exposure to novel sensorimotor environments. Predictors were evaluated in three categories, including: 1) behavioral tests to assess sensory bias and adaptability; 2) imaging to determine individual brain morphological and functional features; and 3) genotype markers for genetic polymorphisms that play a role in the neural pathways underlying sensorimotor adaptation. These two studies were conducted to help characterize the sensorimotor risk profile and design sensorimotor adaptability training countermeasures that may be customized for each crewmember’s individual characteristics. In addition to collecting data from ground subjects, the SM Predictors-Retrospective study gathered data from six crewmembers, and from three subjects who participated in a bed rest campaign in the past. While these data supported the validity of the tested measures, the small number of subjects in this part of the study limited the interpretation of the collected data, and logistical challenges (namely, the end of the NSBRI cooperative agreement) precluded recruiting additional astronaut subjects until now. Meeting HRP’s goal of addressing gap SM 24 requires collection of data from additional crewmembers who fit the recruitment criteria for this study (crewmembers for whom functional task performance data sets exists after previous spaceflight missions). This proposed effort will thus build-upon the findings from the previous studies, with the full data set (from both the prior study and the proposed one) used to build predictive models of postflight functional task performance capabilities. Since this study directly builds upon the two prior studies and data needs to be collected seamlessly, it requires the specific expertise that has been developed within the NASA Johnson Space Center Neurosciences Laboratory. Therefore, the work reflects "focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal."
Research Impact/Earth Benefits:	The goal of this study is to identify a set of behavioral, neuroimaging and genetic measures that can be used to better understand the inter-subject variability in early post-flight performance on a set of sensorimotor tasks. We expect that understanding the relationships between biomarkers and post-flight functional task performance will improve our strategy to optimize sensorimotor countermeasures. These same biomarkers may also be useful for understanding individual variability in balance function in the general population, and predicting sensory compensation associated with aging and/or the onset of vestibular disease.

Task Progress:	During this last reporting period, the final three participants completed the study. Thirty International Space Station (ISS) crewmembers were recruited in total, including 12 first-time fliers, 6F, with mission durations lasting 185.5 ± 45.5 days, mean ± std. This included 22 returning on Soyuz and 8 returning on Dragon. The laboratory tests were completed 3.3 ± 3.8 yrs (mean ± std) since post-landing measures were obtained. A major emphasis of this final reporting period was completion of the neuroimaging analysis using an approach previously published (Koppelmans et al., 2022). We utilized the Brain Imaging Data Structure (BIDS) standard for organizing the preflight MRI datasets (Gorgolewski et al., 2016). Given the potential impacts of myelin remodeling on neural function, pipelines from the Human Connectome Project (HCP) were used to extract myelin maps. The FreeSurfer software package was used to obtain cortical thickness and brain volumes. ACAPULCO, a processing pipeline for cerebellar segmentation on T1-weighted images, was used to obtain cerebellar lobule volumes. TRACULA (TRActs Constrained by UnderLying Anatomy) was used for automatic reconstruction of a set of major white-matter pathways from diffusion-weighted MRI data. TRACULA output was combined with the HCP pipelines output to extract myelin values for the corticospinal tract. Finally, ComBat was used as a harmonization technique to account for multiple software versions / calibrations of the neuroimaging test center (Victory Lakes) while preserving the inter-subject variability that is the basis of this study. While the results of our study observed specific relationships between ground behavioral biomarkers and post-flight outcome measures, we did not find a pattern of individual sensory biases or adaptive capabilities that predicted overall functional performance after long-duration spaceflight. This was limited since performance on one post-flight test does not necessarily correlate with performance on other tests. While some of the preflight brain structure measures also predicted specific postflight balance outcomes, we did not detect patterns in polymorphisms in a limited set of genetic biomarkers (COMT, DRD2, BDNF, and α2-adrenergic receptor) to be associated with postflight outcomes. Subject demographics (e.g., flight experience, sex, landing vehicle) were also not significantly related to postflight outcomes. Outcome measures from prior flights may be helpful in predicting similar outcomes on future flight opportunities, although posturography measures across missions in this cohort were highly variable. We conclude that future work should focus on validating post-landing assessments that can predict operational task performance (e.g., assessments performed prior to early lunar surface activities).
Bibliography Type:	Description: (Last Updated: 02/20/2026)
Abstracts for Journals and Proceedings	Wood SJ. "The diversity of adaptative responses following g-transitions during spaceflight." Vestibular-Oriented Research Meeting, Boulder Colorado, May 19-22, 2025. J Vestibular Res, Volume 35, Issue 5, https://doi.org/10.1177/09574271251347380 , May-2025
Abstracts for Journals and Proceedings	Wood SJ. "Postural recovery following spaceflight." XXXII Bárány Society Meeting, Uppsala, Sweden, August 25-28, 2024. J Vestibular Res, Volume 34, Issues 5-6. , Aug-2024
Abstracts for Journals and Proceedings	Wood SJ. "A review of sensorimotor readaptation following spaceflight." Vestibular-Oriented Research Meeting, Boulder Colorado, June 26-29, 2023. J. Vestibular Res, Volume 33, Issue 4, https://doi.org/10.3233/VES-230300 , Jun-2023
Abstracts for Journals and Proceedings	Wood SJ, Moudy SC, Macaulay TR, Bishop MO, Clément GR. "Field assessment of sensorimotor function following long-duration spaceflight." 93rd Annual Scientific Meeting of the Aerospace Medical Association, New Orleans, Louisiana, May 21-25, 2023. Aerospace Med Hum Perform Vol. 94, No. 4 April 2023. , May-2023
Abstracts for Journals and Proceedings	Wood SJ, De Dios YE, Macaulay TR, Peters BT, Beltran NE, Koppelmans V, Clark TK, Seidler RD, Oddsson L, Theriot CA, Bloomberg JJ. "Assessing the relationships between sensorimotor biomarkers and post-landing functional task performance." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Tays GD, Koppelmans V, De Dios YE, Seidler RD, Macaulay TR, Peters BT, Clark TK, Oddsson L, Theriot CA, Bloomberg JJ, Wood SJ. "Assessing the relationships between sensorimotor biomarkers and post-landing functional task performance." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
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